

# Post-quantum transition for device bounded anonymous credentials: a EUDI wallet application

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## Abstract

In 2026, member states of the European Union are expected to have running their own EUDI wallet applications instances compliant with the ARF architecture guidelines. The goal of the ARF architecture is to provide privacy-preserving anonymous credentials to the users, but the current implementation fails. In the meantime, in 2024 the European Union has published a roadmap for the post quantum digital transition of members digital infrastructures.

No current implementation of anonymous credentials is post quantum secure at the moment, with cryptologist and mathematician in top publications claiming that is not possible to build a post quantum anonymous credential scheme with the actual technologies, given hardware constraints and advanced requirements. So we asked ourselves: is it possible to build a post quantum device bounded anonymous credential scheme with the actual technologies, giving hardware constraints and technical difficulties? The answer is **yes**, it is possible.

Moreover, in this thesis we present a post quantum anonymous credentials model that can be deployed with the actual technologies and legacy infrastructures, with NIST standardized post quantum signature schemes. We went further and implemented the full model and run it into android devices in order to benchmark the performance and the results are surprising. We also present some optimizations that could be implemented in order to further improve the performance of the scheme, and we show how to implement them with the actual technologies.

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# 1 Introduction

Anonymous credentials are a powerful tool for preserving privacy in digital interactions between users and service providers. Anonymous credentials were introduced by David Chaum in 1982 [6], and later formalized by Camenisch and Lysyanskaya in 2001 [3, 4]. They allow users to prove their identity or attributes without revealing any additional information about themselves.

The usual model of anonymous credentials consists of three parties:

- The issuer, a central entity who is responsible for issuing credentials to users and eventually revoking them;
- The user, who asks credentials to the issuer and holds them, and eventually use them to prove their identity or attributes to verifiers.
- The verifier, who is responsible for verifying the user’s credentials without learning any additional information about the user.

A basic scheme of how this entity interacts with each other is represented in figure 1.

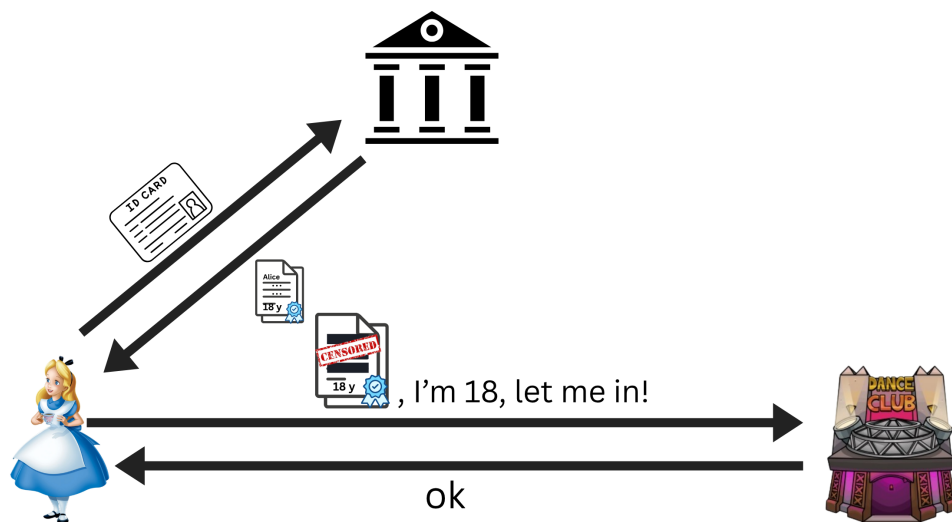


Figure 1: Basic scheme of anonymous credentials.

In the first approach [13], the issuer was signing user’s credentials using a secure protocol that protects user’s attributes. Credential(s) are then showed to a verifier via zero-knowledge proof of knowledge of a credential(s) whose attributes satisfy some predicate.

Systems based on anonymous credentials have been studied and adopted over the years by big tech companies like Microsoft [15], IBM [5], and are in continuous study and development by the scientific community, facing challenges like performance improvements, revocation lists, legacy infrastructures and hardware constraints, and more recently, post quantum security.

## 1.1 Post quantum transition

The european union on 11 April 2024 has published a roadmap for the post quantum digital transition of members digital infrastructures [9].

The roadmap is a comprehensive plan that outlines the steps that the European Union will take to transition to post-quantum cryptography. The roadmap includes a timeline for the transition, as well as a list of actions that will be taken to ensure a smooth transition.

The timeline is organized into three major phases:

1. **By the end of 2026:** Initial national PQC transition roadmaps have been established by all Member States, PQC transition planning and pilots for high- and medium-risk use cases have been initiated;
2. **By the end of 2030:** The PQC transition for high-risk use cases has been completed, PQC transition planning and pilots for medium-risk use cases have been completed. Quantum-safe software and firmware upgrades are enabled by default.
3. **By the end of 2035:** The PQC transition for medium-risk use cases has been completed. The PQC transition for low-risk use cases has been completed as much as feasible.

As previously said, this roadmap has been published by the European Commission in 2024. The recent advances in the field of post-quantum computing due to hardware improvements, artificial intelligence acceleration and advances in research have made the Q-day closer than expected, and the transition to post-quantum cryptography more urgent.

On 30 March 2026, Google published a drastical advance in post-quantum cryptography: they reduced consistently the amount of time, resources and qbits needed to break elliptic curves based cryptography [2]. In the meanwhile, between April and March 2026, big tech companies like Google [1] and Cloudflare [18] have announced that they will accelerate the transition to post-quantum cryptography, lowering the timeline for the full transition to 2029, instead of 2035.

## 1.2 Overview

In this thesis, we present a post-quantum anonymous credentials model that can be deployed with the actual technologies and legacy protocols. We require the issuer to adopt ML-DSA NIST standardized post quantum signature scheme, and to slightly leverage the credentials structure in order to accommodate the larger size of post quantum signatures and public keys, as well as modification to the mdoc structure in order to involve a commitment of the device public key instead of the device public key itself.

In the following sections, we are going to present an overview of the EUDI wallet 2, we perform a full analysis of the actual state of the art on device bounded anonymous credentials 3, then we explain how our solution faces and solves the challenges, performing an important advance on the actual state of the art 4. We show how we implemented our solution, giving a reference to our github repository 5, and we present our results 6, future works that we plan to do 7 and conclusion 8.

# 2 European Digital Identity Wallet

In 2021, the European Commission proposed to adopt a framework for a European Digital Identity [7]. In 2024, the European Commission decided to establish an Architecture Reference Framework (ARF) [8], which consisted of a set of guidelines and properties that each instance of the wallet application should satisfy. This general framework is needed since the commission left freedom to each member state to implement their own instance of the application, due to different possible credentials structures (for example, french and german national identity cards have different structures and/or attributes) and infrastructures (for example, some member states have already a national identity card with an embedded chip, while others don't).

In the following section, we are going to analyze the architecture of the EUDI wallet application, to give a complete overview of Architecture Reference Framework, and to analyze the vulnerabilities of the proposed implementation.

## 2.1 Cryptographic foundations

The entire ecosystem is built on a **Public Key Infrastructure (PKI)**. The main primitives and standards are the following:

- **Elliptic-Curve Cryptography (ECC)** is the preferred choice for mobile-device performance. The relevant presentation standards, ISO/IEC 18013-5 [12] and SD-JWT VC [11], rely respectively on the **COSE** (CBOR Object Signing and Encryption) and **JOSE** (JSON Object Signing and Encryption) frameworks.
- **RSA** is also supported, together with **X.509** certificates [?] and the ETSI standards used for qualified electronic signatures.

For signing documents and transactions the ARF explicitly references the ETSI *AdES* (Advanced Electronic Signature) profiles, namely **XAdES**, **PAdES**, **CAdES** and **JAdES**, depending on the data format being signed.

It is important to stress already at this point that, although the PKI is format-agnostic in principle, the *device-binding* keys – the keys that anchor a credential to a specific device – must reside in the device’s secure hardware, which in practice today only supports ECDSA over the NIST P-256 curve. This constraint is central to ??.

## 2.2 Functionalities

At a high level, the functionalities offered by a Wallet Unit can be grouped into four categories:

- **Secure identification and authentication**, ensuring that users can present person identification data (PID) in a trusted environment, both online and in proximity.
- **Exchanging qualified and non-qualified user attributes** through secure and verifiable electronic attestations of attributes (EAAs).
- **Electronic signing of documents or data**, allowing users to create legally recognised qualified electronic signatures and seals.
- **Generation and use of pseudonyms** for authentication, to enhance privacy and prevent tracking.

A further, emerging use case is the wallet acting as a *strong customer authentication* device for payments, discussed in section 2.2.5.

Now we will discuss in depth each of these functionalities, describing the involved actors and the protocols used to implement them.

### 2.2.1 Identification and authentication

**Description.** Users can identify and authenticate to online and offline services while using *selective disclosure* of attributes or attestations.

**Actors.** User; Wallet Unit (a smartphone application); Relying Party (the service requesting data); PID Provider.

**Protocols.** Two complementary protocols are used:

- **ISO/IEC 18013-5** [12] for *proximity* flows. The Wallet Unit and the Relying Party set up a communication channel using a QR code or NFC and then communicate over BLE, NFC or Wi-Fi Aware. The standard provides confidentiality and authenticity for all exchanged data, as well as Relying Party authentication. Figure 2 sketches the proximity flow.
- **OpenID4VP** [14] for *remote* (including cross-device) flows. It defines the message structures, transaction flows and an HTTP-based interface for presenting attestations from a Wallet Unit to a Relying Party, again providing confidentiality, authenticity and Relying Party authentication. Figure 3 sketches the remote, cross-device flow.

In both protocols, **device binding** is achieved by including a cryptographic public key in the PID or attestation and signing it; the corresponding private key is protected by a Wallet Secure Cryptographic Application/Device (WSCA/WSCD) or keystore inside the Wallet Unit. Although the proximity and remote protocols differ in their details, at a high level they both implement **Relying Party authentication** as illustrated in fig. 4.

### 2.2.2 Exchanging user attributes

**Description.** The wallet lets a user request, store and share Electronic Attestations of Attributes – EAA, PuB-EAA and QEAA. The wallet supports *selective disclosure*, so that the user can share only a subset of the attributes contained in an attestation.

**Actors.** User; Wallet Unit; Relying Party; Attestation Provider (QEAA, PuB-EAA or EAA), the entity issuing the attestation.

**Protocol.** *SD-JWT VC* [11] (Selectively Disclosable JSON Web Token Verifiable Credentials) is a special form of JWT that is selectively disclosable. It specifies the JSON encoding of attributes and metadata together with rules that prevent collisions of claim names; a proof mechanism that ensures the authenticity and integrity of a PID or attestation while allowing selective disclosure of individual attributes; and an *optional* security mechanism enabling device binding of PIDs and attestations. Figure 5 illustrates the structure of an SD-JWT VC.

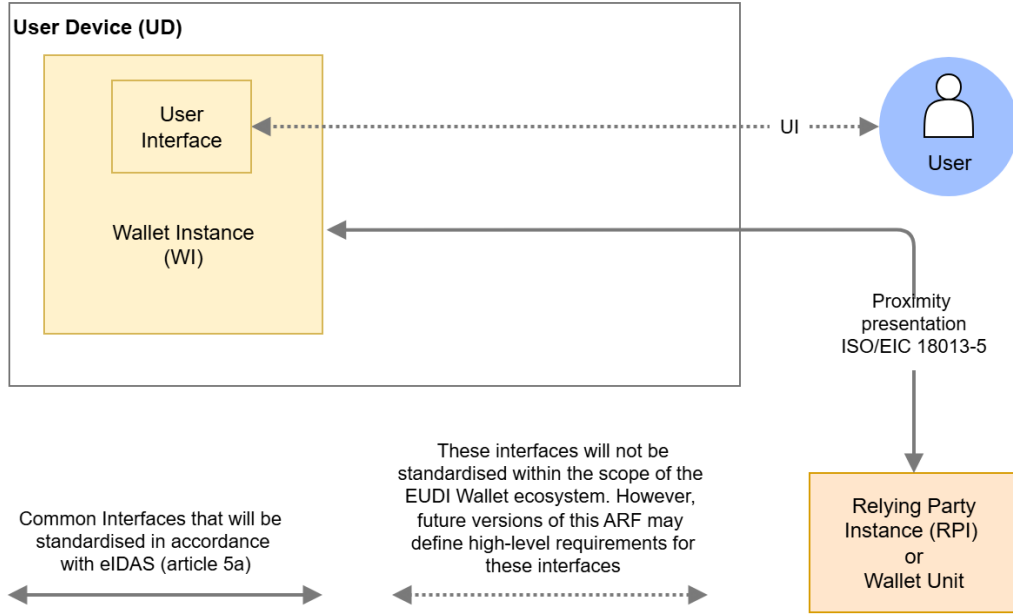


Figure 2: Proximity presentation flow (ISO/IEC 18013-5). Adapted from the ARF [8].

**How attestations are added.** For mobile driving licences (mDL) the Attestation Rulebook explicitly references the ISO/IEC 18013-5 model, the same used for identification. Other attestations are obtained as follows: the user first verifies that the provider is trusted (it appears on an official Trusted List or List of Trusted Entities); the provider authenticates the user, typically reusing the OpenID4VP authentication flow; the provider validates the Wallet Instance through its Wallet Instance Attestation; the user reviews and approves the attestation; and finally the wallet verifies the provider’s signature on the attestation and stores it.

**How attestations are stored.** The ARF mandates a small set of standardised formats:

- **CBOR**, used by ISO/IEC 18013-5. It is mandatory for mDLs and may be used for any other attestation if the provider remains compliant with the standard.
- **JWT**, used by SD-JWT VC. No credential is required to be expressed in this format, but wallet support is mandatory.
- **W3C Verifiable Credentials Data Model v2.0** [16], an optional format valid only for non-qualified EAAs and still under development.

Table 1 summarises the three formats and their proof mechanisms.

| Table 1: Attestation formats and proof mechanisms supported by the ARF. |                           |                             |                              |
|-------------------------------------------------------------------------|---------------------------|-----------------------------|------------------------------|
| Feature                                                                 | ISO/IEC 18013-5 / 23220-2 | SD-JWT VC                   | W3C VCDM v2.0                |
| Data format                                                             | CBOR (binary)             | JSON Web Token (JSON)       | JSON-LD (linked data)        |
| Proof type                                                              | Embedded (salted hashes)  | Embedded (salted hashes)    | Detached or embedded         |
| Key use case                                                            | Proximity (e.g. mDL)      | Remote (e.g. remote ident.) | General (requires profiling) |
| Wallet support                                                          | Mandatory                 | Mandatory                   | Optional                     |

**How selective disclosure works.** Both ISO/IEC 18013-5 and SD-JWT VC realise selective disclosure through the *salted-hash* technique combined with device and user binding:

- **Salted hashes.** The issuer signs a token in which each attribute is replaced by a salted hash of its value rather than its clear text (e.g. the hash of `family_name: Rossi` together with a random salt). To disclose an attribute, the user reveals the clear value and the salt; the verifier recomputes the hash and checks that it matches the one inside the signed token.

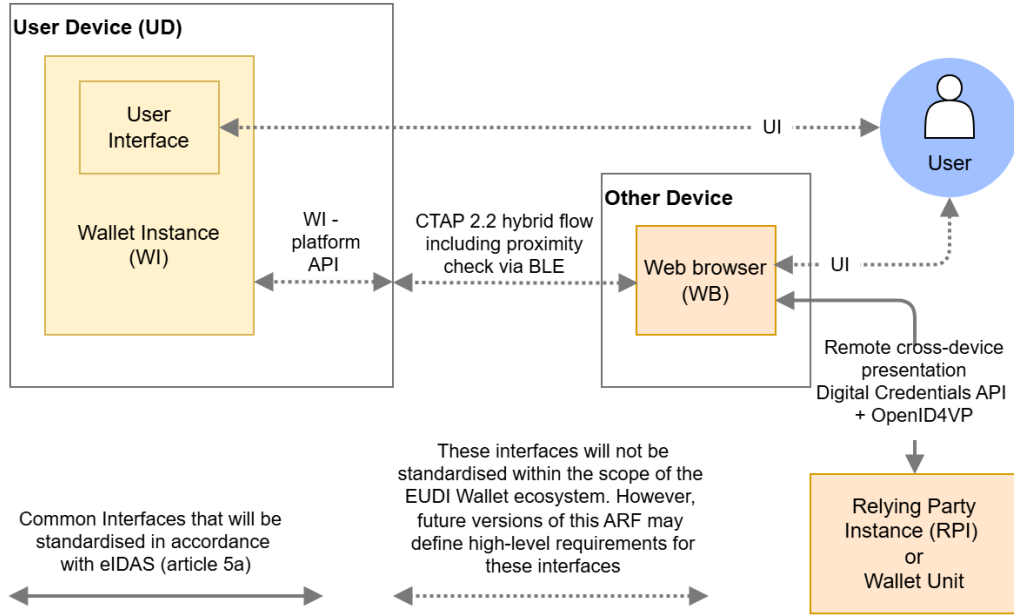


Figure 3: Remote, cross-device presentation flow (OpenID4VP). Adapted from the ARF [8].

- **Device binding.** The attestation embeds a public key whose private counterpart never leaves the WSCD. The provider/verifier issues a random challenge that the device signs, proving that it holds the private key bound to the attestation.
- **User binding.** Before disclosure, the wallet shows the user which attributes are about to be sent and requires local authentication (PIN, biometrics, etc.).

Zero-knowledge proofs, which would allow attributes to be proven without revealing the underlying field, are not yet part of the supported mechanisms; their status is discussed later in the analysis.

### 2.2.3 Qualified electronic signatures (QES)

**Description.** The wallet lets users create legally recognised qualified electronic signatures and seals for legally binding transactions and document signing.

**Actors.** User; Wallet Unit; a *local QSCD* (a local tamper-resistant device that produces signatures) or a remote *QESRC* provider (Qualified Electronic Signature Remote Creation).

**Protocol.** Communication between the Wallet Unit and the QSCD/WSCD takes place through a *Secure Cryptographic Interface*. All parties in the ecosystem that handle cryptographic keys – Wallet Providers, PID and Attestation Providers, Trusted List/LoTE Providers, Providers of Registration Certificates and Access Certificate Authorities – typically rely on a certified **Hardware Security Module (HSM)** to generate and manage private and secret keys. In other words, the security of this functionality rests almost entirely on *hardware* components, which perform key generation and management, user authentication, and the production of signatures and seals; the choice of the actual key-generation protocol is delegated to the wallet implementer.

### 2.2.4 Pseudonyms

**Description.** Users can present a pseudonym to a Relying Party in situations where the Relying Party does not need to learn the user’s identity, which helps users avoid being tracked across services.

**Actors.** User; Wallet Unit; Relying Parties.

**Protocol.** The ARF distinguishes three kinds of pseudonyms:

- *Verifiable* pseudonyms, which let the user prove ownership;
- *Attested* pseudonyms, a sub-type of verifiable pseudonyms in which a third party attests that the pseudonym belongs to the user;
- *Scope rate-limited* pseudonyms, a sub-type that guarantees the user controls at most a fixed number (the *rate*) of pseudonyms for a given scope.

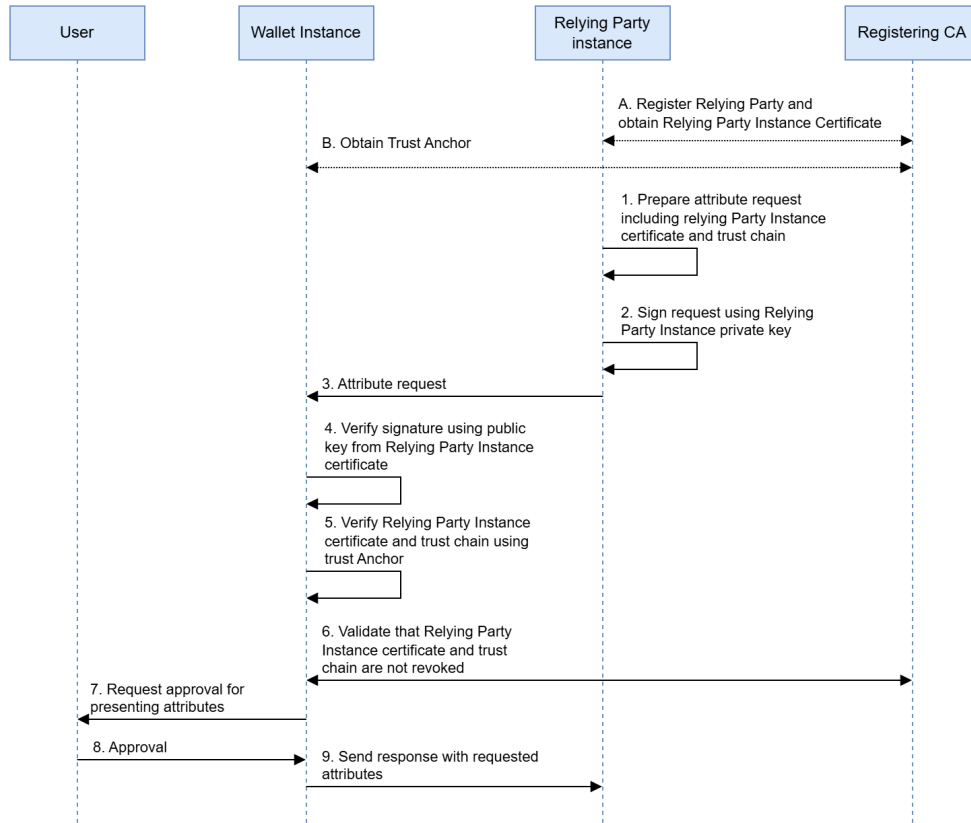


Figure 4: Relying Party authentication, common to the proximity and remote flows. Adapted from the ARF [8].

**W3C WebAuthn** [17] specifies a class of verifiable pseudonyms known as *passkeys*. The protocol has two phases:

1. **Registration.** The user generates a public/private key pair, stores both keys in a secure device (the *Authenticator*), and registers the public key with the desired Relying Party.
2. **Authentication.** When the user wishes to authenticate, the service sends a challenge consisting of a random value; the user signs it with the private key held by the Authenticator and returns the signature; the service verifies the signature against the registered public key and, if the signature and the origin match, grants access.

Pseudonyms are stored encrypted in a local secure area of the device. Figure 7 illustrates the WebAuthn authentication scheme.

### 2.2.5 Payments

The EUDI Wallet acts as an *authenticator* that lets the user authorise payments. Unlike, for example, Google Wallet – which typically stores a tokenised version of a physical card – the EUDI Wallet does not store a card number. Instead, it is issued a **Strong User Authentication (SUA)** attestation: a verified digital document that links the Wallet Unit to a specific bank account or payment service. This SUA attestation is *device-bound*, i.e. cryptographically tied to the secure hardware (WSCD/WSCA) of the user’s phone.

When a payment is made, the merchant (acting as a Relying Party) sends the wallet a request containing the transactional data. The user reviews the request on screen and, upon consent, the wallet signs that specific data with its internal keys to authorise the payment. The process is designed to satisfy the **Strong Customer Authentication (SCA)** requirements of the EU payment-services framework (PSD2/PSD3) [10]. The payment lifecycle therefore has three phases: *registration* (linking the wallet to the bank via an SUA attestation), *authentication* (approving a transaction by signing the amount and payee) and *termination* (unlinking by revoking the SUA attestation).

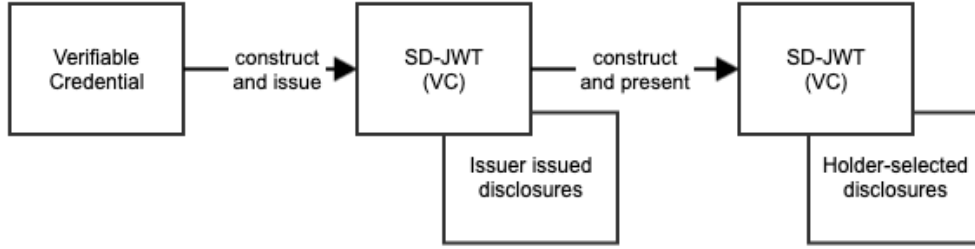


Figure 5: Structure and selective-disclosure mechanism of an SD-JWT VC. Adapted from the SD-JWT VC specification [11].

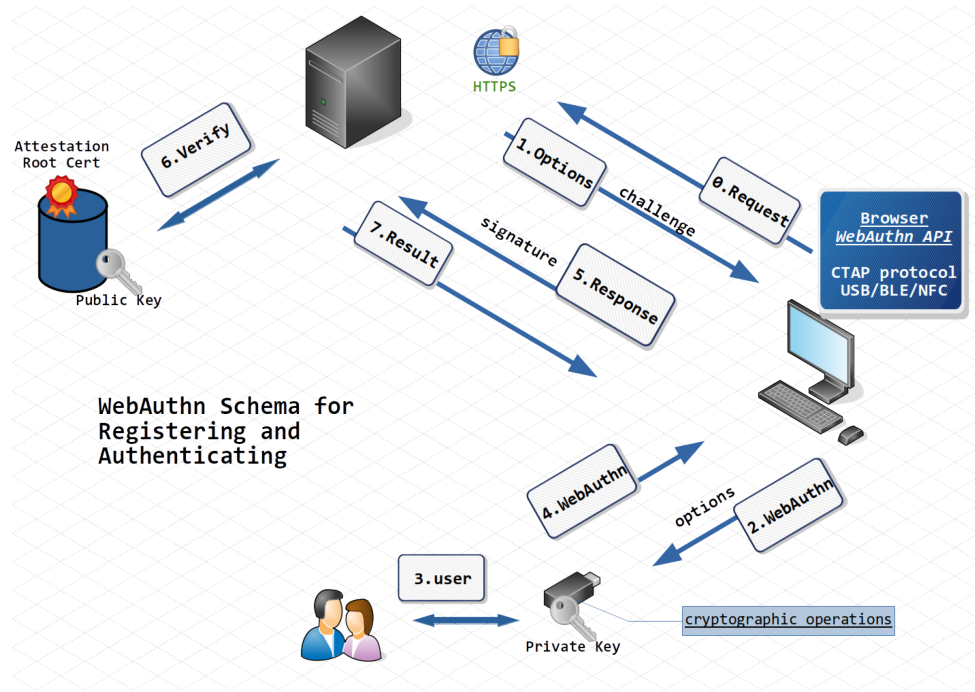


Figure 6: W3C WebAuthn authentication scheme used for verifiable pseudonyms (passkeys). Adapted from [17].

## 2.3 Roles



| Role                               | Primary Responsibility                                                        | Section          |
|------------------------------------|-------------------------------------------------------------------------------|------------------|
| <b>User of Wallet Unit</b>         | Manage, store, and present PIDs/attestations.                                 | ARF Section 3.2  |
| <b>Wallet Provider</b>             | Make the certified Wallet Solution available to Users.                        | ARF Section 3.3  |
| <b>PID Provider</b>                | Issue Person Identification Data (PID) to Users.                              | ARF Section 3.4  |
| <b>Trusted List/LoTE Provider</b>  | Maintain, manage, and publish Trusted Lists and/or Lists of Trusted Entities. | ARF Section 3.5  |
| <b>QEAA Provider</b>               | Issue Qualified Electronic Attestations of Attributes (QEAs).                 | ARF Section 3.6  |
| <b>PuB-EAA Provider</b>            | Issue EAs on behalf of a public sector body.                                  | ARF Section 3.7  |
| <b>EAA Provider</b>                | Issue Non-Qualified Electronic Attestations of Attributes (EAs).              | ARF Section 3.8  |
| <b>QESRC Provider</b>              | Provide Qualified Electronic Signature Remote Creation services.              | ARF Section 3.9  |
| <b>Authentic Source</b>            | Act as the definitive repository for specific attributes.                     | ARF Section 3.10 |
| <b>Relying Party (RP)</b>          | Request and receive attributes from a Wallet Unit.                            | ARF Section 3.11 |
| <b>CAB</b>                         | Certify Wallet Solutions and audit Trust Service Providers.                   | ARF Section 3.12 |
| <b>Supervisory Body</b>            | Review the proper functioning of ecosystem actors.                            | ARF Section 3.13 |
| <b>Device Manufacturers</b>        | Provide the underlying platform (hardware, OS, secure elements).              | ARF Section 3.14 |
| <b>Attestation Scheme Provider</b> | Define and publish the Attestation Rulebooks and schemes.                     | ARF Section 3.15 |
| <b>NAB</b>                         | Accredit CABs according to EU regulations.                                    | ARF Section 3.16 |
| <b>Registrar</b>                   | Manages the registration of Providers and Relying Parties.                    | ARF Section 3.17 |
| <b>Access CA</b>                   | Issue access certificates for authentication.                                 | ARF Section 3.18 |
| <b>Provider of Reg. Certs</b>      | Issue certificates detailing registration status and scope.                   | ARF Section 3.19 |



## 2.4 Zero-knowledge proofs in the ARF

Citing ARF developer guide at section 7.4.3.5.3 Zero-knowledge proofs, Unlike Relying Party linkability, attestation Provider linkability cannot be fully eliminated when using attestation formats based on salted hashes. The only viable mitigation, they say, is to adopt Zero-Knowledge Proofs (ZKPs) as a verification mechanism instead of relying on salted-attribute hashes.

Furthermore, navigating the ARF developers guide, we found the following statements: Zero-Knowledge Proof (ZKP) mechanisms for verifying personal information are highly promising and essential for ensuring privacy in various use cases. They enable Users to prove statements such as "I am over 18" without disclosing any personal data, offering a robust solution for privacy-preserving authentication and verification.

However, the integration of ZKPs in the EUDI Wallet ecosystem is still under discussion and development, due to the complexity of implementing ZKP solutions in secure hardware and the lack of support in currently available secure hardware (WSCDs). As with Relying Party linkability, organisational and enforcement measures can help deter Attestation Providers from colluding and tracking Users. Additionally, many Attestation Providers are subject to regular audits, making it easier to detect collusion and tracking compared to Relying Parties.

## 2.5 Vulnerabilities

# 3 State of the art/Literture overview on device bounded anonymous credentials

## 3.1 Anonymous Credentials

## 3.2 Post quantum anonymous credentials

# 4 Our solution

## 4.1 Claim

## 4.2 Analysis

### 4.2.1 Security analysis

**MAC witness sharing** STARK zk proof over the issuer signature needs dilithium verification algorithm. The algorithm takes in input the signature  $sig$ , the message being signed  $e_1$  and the public key of the issuer  $PK_{II}$ . The message signed by the issuer  $e_1$  is the sha256 of the credential document. Knowledge of sha256 preimage is hard to prove in STARK winterfell circuit, so we propose the following solution: the Liger circuit, which has an optimized sha prover, will prove  $e_1 = sha256(cred)$ , then  $e_1$  is encoded in a binding and hiding commitment realized with information theoretic MAC. ecc ecc ecc. Mathematical proof of MAC security:

$$m = MAC_k(e_1) = MAC_{k_p, k_v}(e_1) = MAC_{a, b}(e_1)$$

We obtain the key as:

$$(a, b) = k_p \oplus k_v$$

where  $k_p$  is a secret random 32 bytes key chosen by the prover and  $k_v$  is a public verifiable key chosen by the verifier. What security concerns do we have?

1. The prover can try to forge a valid MAC for a different  $e'_1$ , but  $e'_1$  would still need to be a valid sha256 of signed credential since the signature is verified by the circuit;
2. The prover could generate  $k_p$  after seeing  $k_v$ , leading to arbitrary proves;
3. Different presentations could be linkable to the same MAC.

2 is avoidable if we require the prover to generate  $k_p$  before seeing  $k_v$ , committing to and building  $k_v$  with fiat shamir transform on  $k_p$  commitment, and 3 is avoidable if we require the prover to generate a different  $k_p$  for each presentation.

For 1, we have the following security proof in which we show that is not possible to forge a valid MAC for a different  $e'_1$  without breaking the security of the MAC scheme:

$$m = MAC_k(e_1) = a * e_1 + b = a * e'_1 + b \Rightarrow a * (e_1 - e'_1) = 0$$

which is true either if  $a = 0$  or if  $e_1 = e'_1$ . It has been proven that the probability that  $a = 0$  is negligible. If  $e_1 = e'_1$  then the prover is proving the same credential, so no forgery is happening. We also now that is not possible to have  $m_1 = a' * e_1 + b' = m_2 = a'' * e'_1 + b''$  since the prover is not able to choose  $a$  and  $b$  due to the verifier participation in the key generation with  $k_v$ .

#### 4.2.2 Compliance with eIDAS 2.0 and LoA High

In this section we are going to analyze our solution, taking into account the requirements of the European Commission for the EUDI wallet application, considering both the eIDAS 2.0 regulation and the LoA High (Level of Assurance High).

#### Dynamic Authentication and Replay Protection

CIR 2015/1502 requires that for LoA High, authentication must be dynamic, creating a proof that changes with each session. The plan fulfills this by including a session transcript ( $tr$ ) in the Device Binding proof. The device's Secure Element signs this transcript, and the ZK circuit verifies this signature ( $p256.verify$ ), ensuring the proof cannot be replayed by an attacker who has intercepted a previous proof.

#### Hardware Binding

The plan enforces Device Binding by extracting the public key ( $pk_d$ ) from the signed MSO and requiring a proof of possession of the corresponding private key. This ensures the credential is bound to a tamper-resistant WSCD (Wallet Secure Cryptographic Device).

#### Algorithmic Soundness

The extraction error of the combined STARK/Ligero system is negligible ( $O(\eta + d \cdot \log(w) / |F|)$ ), providing a robust mathematical barrier against forgery.

#### Compliance with EUDI ARF Requirements

The implementation satisfies several High-Level Requirements (HLRs) from the EUDI Architecture and Reference Framework:

- **User Sole Control:** The ZKP generation happens entirely on the user's device (orchestrated by the Wallet Unit), ensuring the user has sole control over attribute disclosure;
- **Validity Verification:** The circuit enforces validity checks ( $t_{start} < t_{now} < t_{end}$ ) without revealing the actual issuance or expiration dates, as required for privacy-preserving interactions;

## 5 Implementation

## 6 Results

## 7 Future work

## 8 Conclusion

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